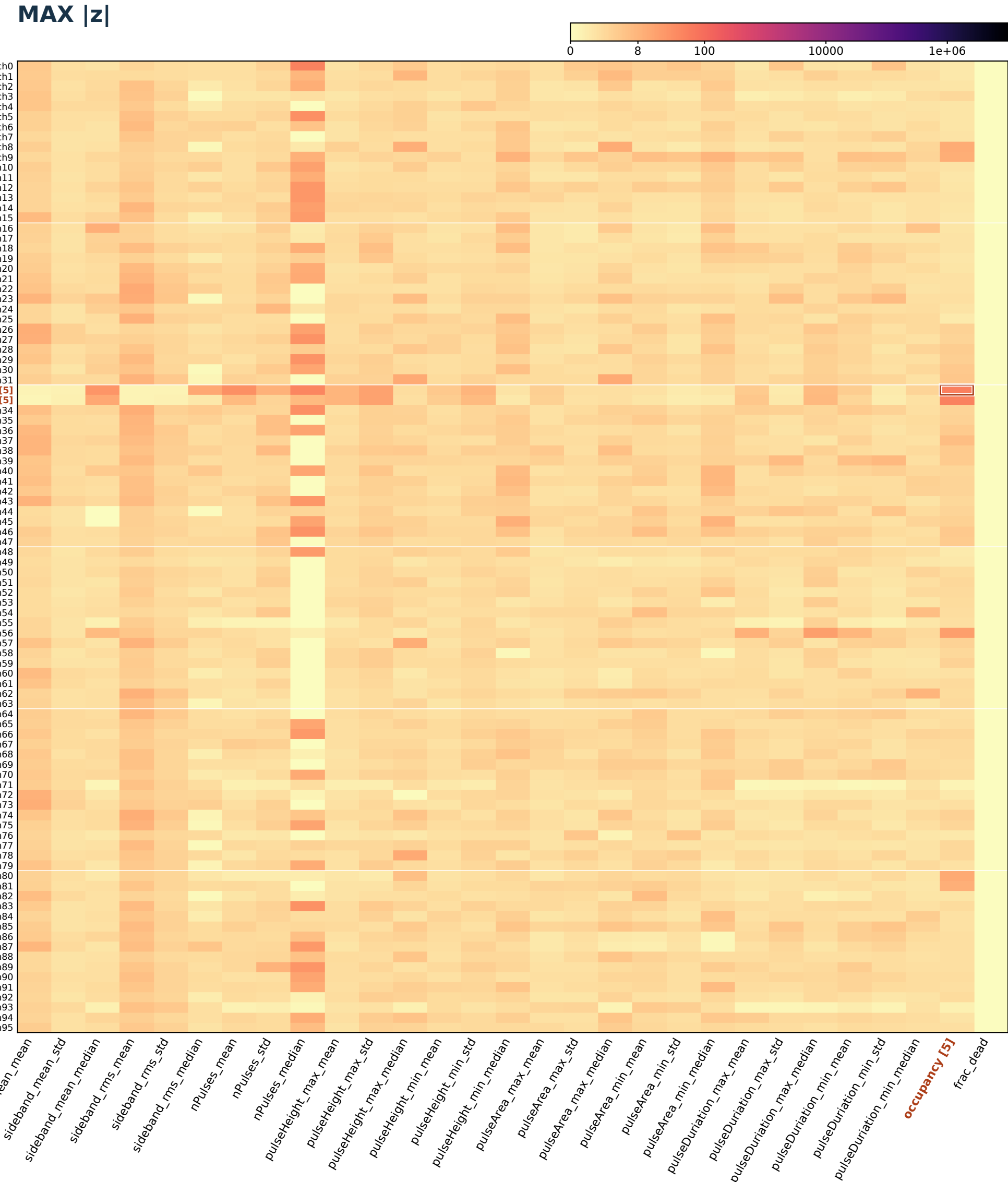


Run 1640 | Observed Signature and LLM-Mentioned Evidence

Frozen IF reference | 96 channels x 29 features | 741/741 valid detector subruns | strict threshold $|z| > 8$ | 5 saved Sol trials

Matrix: digitizer-only model; original feature order and settings. Orange label [n] = explicit text mention in n/5 trials, not internal model attribution.



Max values are NOT clipped; nonlinear color scale resolves both small and million-scale z . Frequency uses each cell's own valid-subrun denominator.

Grey = undefined / masked / no data, not zero. Missing channels are excluded from cell denominators and separately counted below. Orange cell borders require an explicit channel-feature link.

OBSERVED MATRIX

741/741 subruns have defined z values; 0 missing/empty/no- z .

3 cells exceed $|z| > 8$ in at least 80% of their valid subruns.

Most persistent cells (not a cause ranking):

ch32 / occupancy: 99.8%; max 54.2; n=519.

ch32 / pulseHeight_max_std: 81.7%; max 15.2; n=519.

ch33 / occupancy: 81.1%; max 47.4; n=519.

ch33 / pulseHeight_max_std: 68.4%; max 15.3; n=519.

Channel absence is separate from z :

ch32: absent in 222/741 detector subruns.

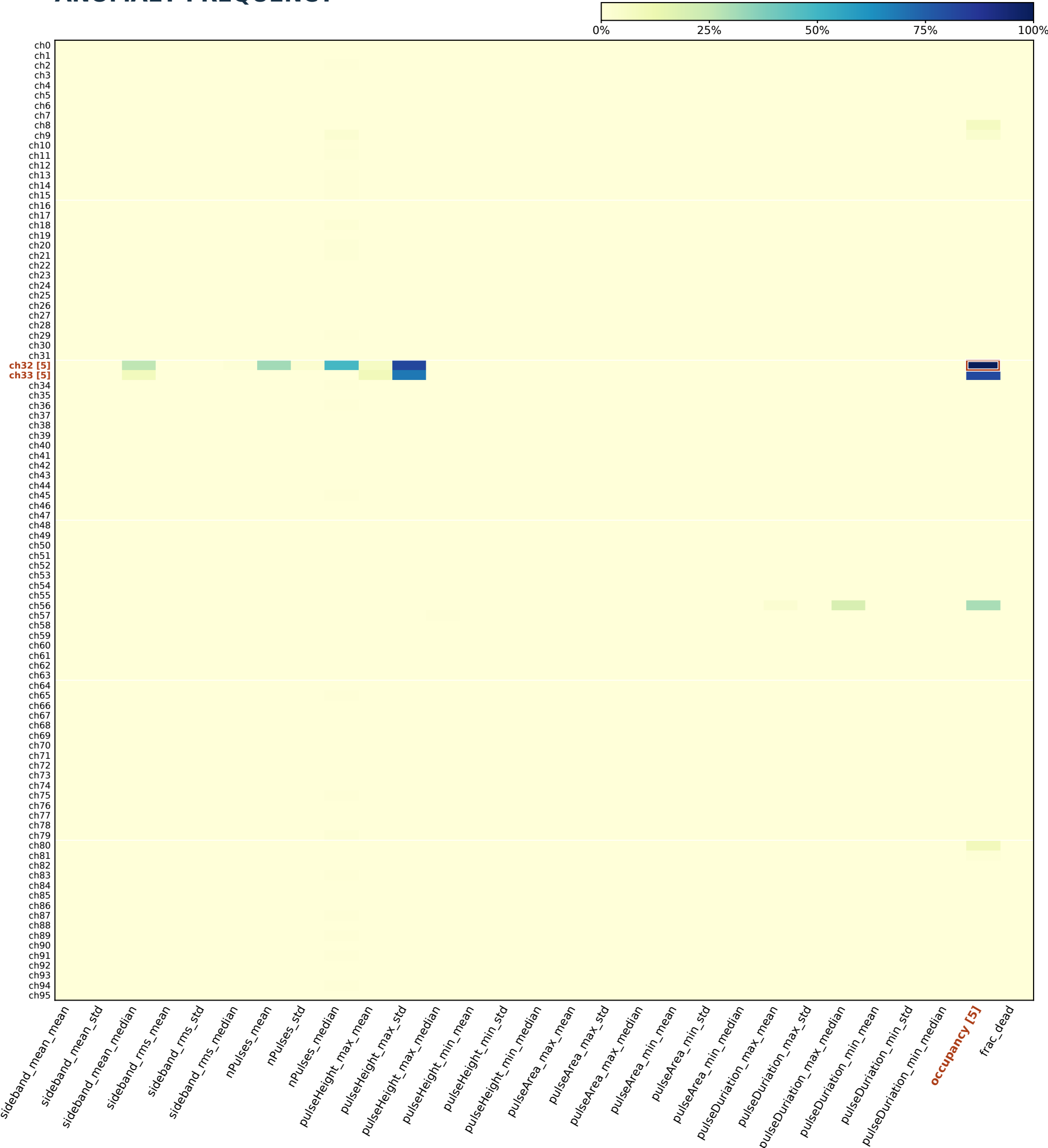
ch33: absent in 222/741 detector subruns.

Frozen IF model has trigger-config masking OFF. Mask text mentions do not establish masked digitizer channels.

LLM-MENTIONED EVIDENCE

Counts are trials, not word occurrences. Text counts include hypotheses/checks; observed-section mentions are shown separately.
configuration: 5/5 text; 1/5 observed-section.
trigger mask: 5/5 text; 5/5 observed-section.
missing-channel behavior: 5/5 text; 5/5 observed-section.
LVDS: 5/5 text; 5/5 observed-section.
board matching: 5/5 text; 1/5 observed-section.
readout failure/dropout: 3/5 text; 0/5 observed-section.
restart (proposed or hypothesized): 3/5 text; 0/5 observed-section.
trigger rate: 2/5 text; 1/5 observed-section.
LVDS pin16: 1/5 text; 0/5 observed-section.
Unambiguous target-observation cell links: 1.
No channel-feature Cartesian expansion. Generic pulse statistics, LVDS pins, mask state and missing-channel text remain notes.

ANOMALY FREQUENCY



PHASE 1 DIAGNOSES (saved category)

Trial 1: localized_digitizer_channel_pair readout/configuration fault

Trial 2: channel_32_33_suppression_or_readout_dropout

Trial 3: intermittent_digitizer_channel_pair_readout_fault

Trial 4: trigger_mask_or_channel_configuration_mismatch

Trial 5: trigger_mask_or_configuration_mismatch

Reading across trials: Trials 1/3 emphasize channel-pair readout faults; trial 2 retains suppression/readout alternatives; trials 4/5 emphasize mask/config mismatch.

Category text is reproduced from saved responses; it is not a true-cause label.